

R16-12DB

Long-distance UHF electronic tag reader

Instruction manual



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1. Introduction

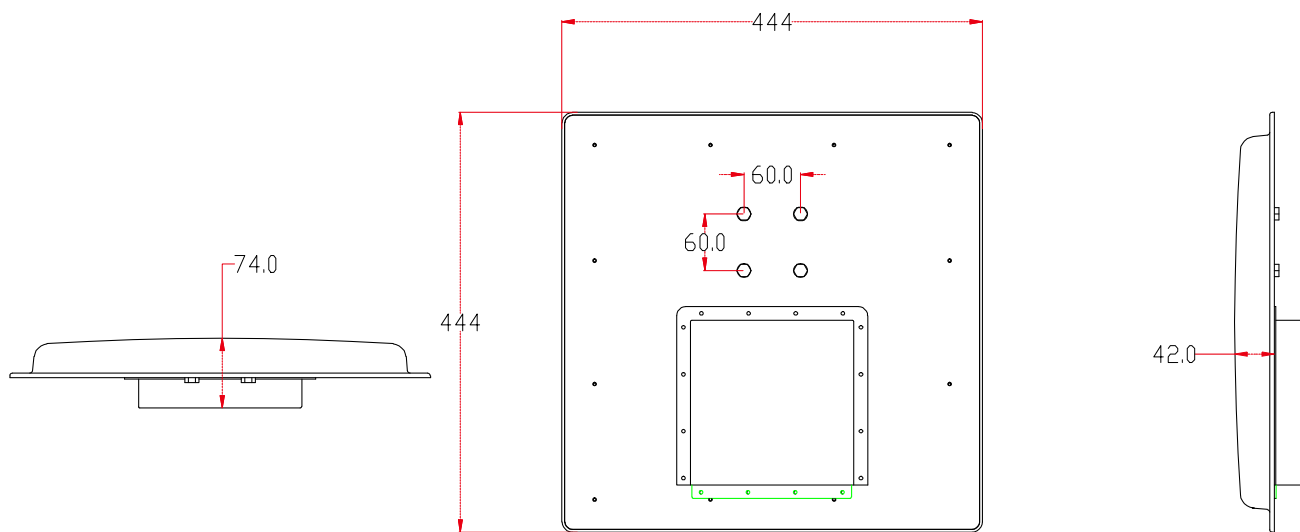
1. This product has multi-protocol compatibility, fast reading rate, waterproof design, to meet the requirements of harsh working environments
2. Full support for ISO-18000-6C (EPC G2) electronic tags;
3. Support USB, RS232 and Wiegand26/34 and other communication methods;
4. The output power is adjustable up to 30dbm, and it supports a variety of working modes such as timing mode, master-slave mode, and trigger mode;
5. The application is suitable for: vehicle access control, automatic charging without stopping, personnel access control management, logistics monitoring, production automation management and other fields

1. Technical parameters

Operating frequency	Standard configuration ISM 902~928MHz and ISM 865~868MHz are available, and other frequency bands can be customized
Antenna gain	Built-in circular polarized antenna with a gain of 12dBi
agreement	ISO18000-6C (EPC G2)
Development kits	Provide SDK development packages, and provide C#, VC, VB, Java, Delphi development routines
Software	Provide test DEMO and automatic card writing, card reading DEMO, convenient for customers to issue and write cards
Output power	0~30dBm, software adjustable
Communication interface	Type N: Wiegand26/34, RS232/RS485, USB Type E: Wiegand26/34, RS232/RS485,TCP, USB
Input interface	1 trigger input
Card reading method	Scheduled automatic card reading and external trigger control card reading, set by the software
Card reading speed	Set by the software, the average reading per 64 bits per 64 bits per card is less than 6ms
Read the distance	0-20m (high power 0 to 35m) (reading distance is related to the label and the site environment).
Operating temperature	-40℃~+65℃
Storage temperature	-45℃~+95℃
Power requirements	DC12V,3A
size	444mm×444mm×74mm
weight	2.55 Kg (with package 4.65kg).

2. The size structure of the reader

Unit: mm



3. Reader use

It can be used for item identification and data acquisition, taking advantage of its good characteristics, especially in the following areas:

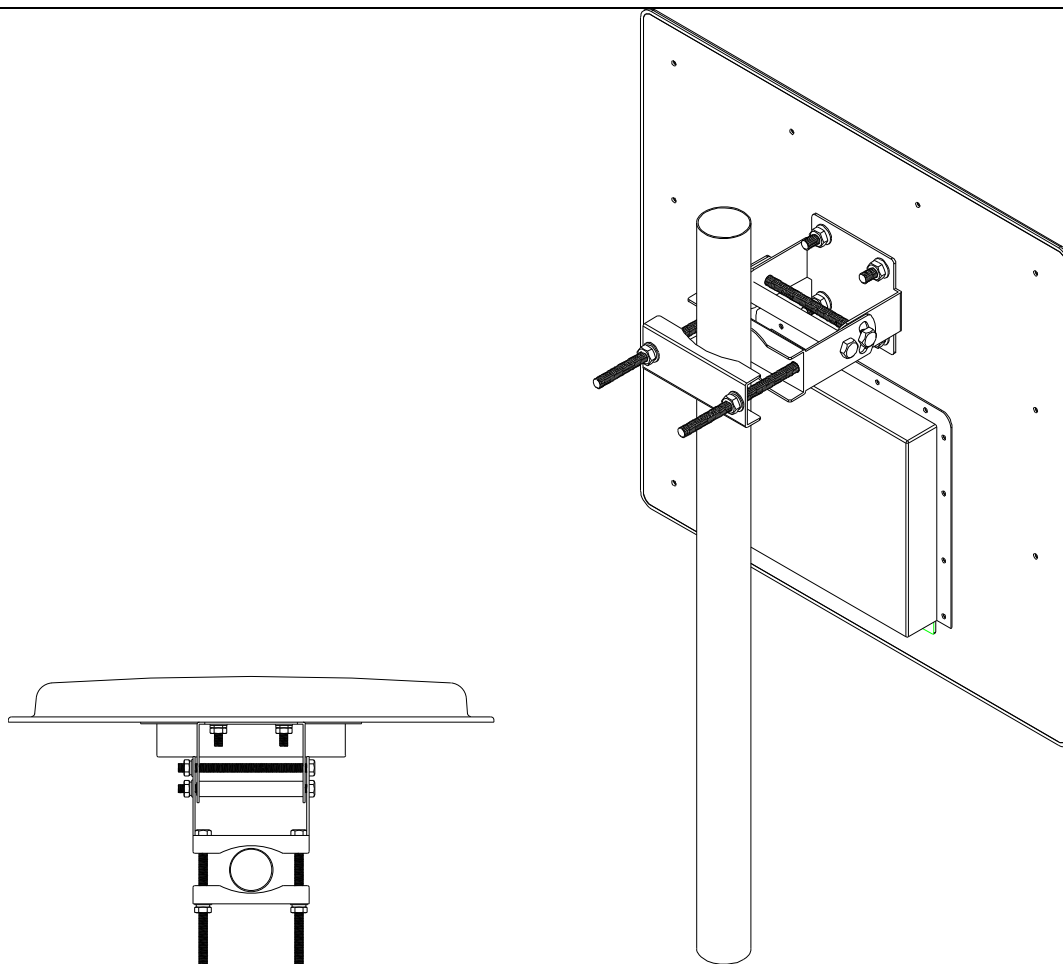
- 1) Transport management: road and rail transport management and container transport management, etc.;
- 2) Motor vehicle management: public security, transportation and other departments of various motor vehicles monitoring and management;
- 3) Road and bridge tolls: Since this product has the ability to read label data at a long distance and at high speed, road and bridge tolls can be carried out without stopping;
- 4) Customs clearance management: customs clearance, customs clearance, customs clearance of materials and vehicles management;
- 5) Warehousing and logistics management: commodity flow and warehousing management, as well as mail, parcels, air baggage, etc. flow management;
- 6) Parking lot management: to automate management and charging;
- 7) Access control management: including the management of the entry and exit of vehicles and

personnel;

8) Process production process: monitor parts throughout the production process;

4. The main function of the reader

- 1) Wake-up tag: Only the awakened tag can communicate with the reader, prevent the interference of other tags outside the system, and ensure the reliability and accuracy of the exchange of tag information between the reader and the system.
- 2) Read tag data: Not only can read the ID number of the tag, but also read the data of the specified tag store; Not only can the data of a single tag be read, but also the data of multiple tags within the antenna wave range can be read simultaneously.
- 3) Write label data: You can write data to a specified label store.
- 4) It can be directly connected to control devices with a standard Wiegand26/34 interface, requiring no development and easy to use.
- 5) Connect with the controller or PC through the standard communication interface for data communication and exchange; Sdk development kits are available for users to further develop applications.
- 6) Precautions for use
 - a) R16 series readers are generally connected to the computer through the RS232 data interface for data exchange. Since the R16 series readers can only read and write electronic tags after receiving the controller's control commands, we will provide customers with SDK development kits that users can use to develop application software.
 - b) The operating temperature of the reader is: $-40^{\circ}\text{C} \sim +65^{\circ}\text{C}$. Therefore, when using this reader in cold areas and seasons, you should pay attention to the pre-heating of the reader 15 minutes before the official use of the reader to ensure the normal operation of the reader.
 - c) It is recommended that when testing, do not have any objects within at least 30 meters in front of the reader; When holding the card, touch your finger on both sides of the card, since this antenna is a horizontal polarization direction, the card should be placed horizontally on the reader during the test to ensure the card reading effect.
 - d) The Wigen communication interface must be connected to the GND ground wire.
 - e) The installation height is more than 1.8M, and the diameter of the pole is 40-80cm.
- 7) Installation diagram



Second, the reader working mode and wiring instructions

1. Wiring instructions for the reader

red	black	green	white	grey	purple
DC+(7-12V)	GND	RX	TX	InG-D1	InG-D 0

2. Label operation

EPC GEN2 (ISO18000-6C) label

- Single-label initialization: Defines the EPC length of the label, typically 96 bits.
- Single tag write: Write to the EPC of the tag, one address or multiple addresses at a time (based on 1 address).
- Single-label lock: Locks the label's EPCs. Once locked, the label's EPCs cannot be overwritten.
- Single Tag Destroy: Destroys the tag. Once destroyed, the label can no longer be used.

3. Working mode

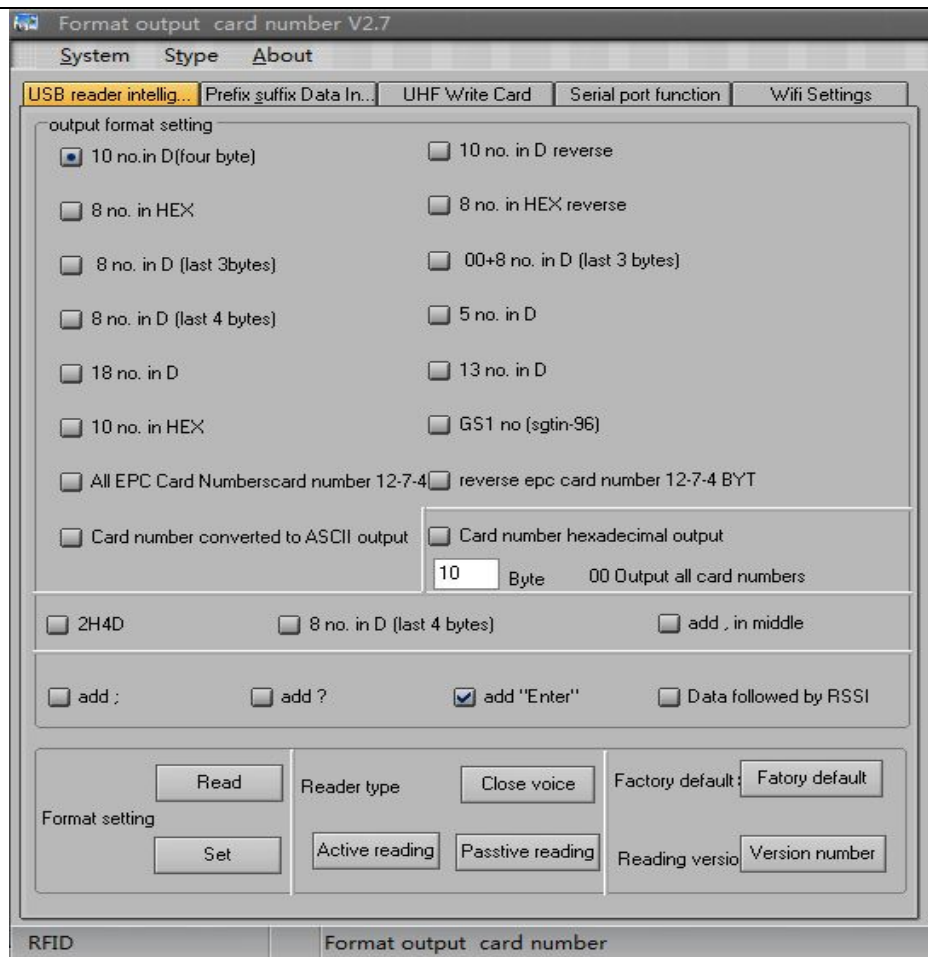
- 1) **Master-Slave Mode of Operation (COMMAND):** In this mode of operation, the reader works under the control of a PC or other controller. The reader and the controller can communicate via RS232, RS485 or one of the Ethernet interfaces. This way of working supports all the features provided by the secondary development kit.
- 2) **Timing READ:** The reader automatically reads the card in a certain period (configurable), and the read data is output through the specified communication port. This mode is read-only for labels.
- 3) **Trigger READ:** When the trigger input port is low, the reader begins to read the card periodically, and automatically shuts down after a period of time.

4. ID adjacency discrimination

The ID neighbor discrimination is designed to reduce redundancy of data uploaded by readers and writers. When this feature is selected, when the reader reads the card number of the same label multiple times in a row, only one set of data is uploaded within a set time. The adjacency discriminant can select the effective time, that is, if the interval between the two adjacent card readings exceeds the effective time, they will not be compared to the adjacency. Users should choose according to their specific needs. Commonly used in Wiegand communications.

Third, the reader configuration

The company provides a DEMO software program for configuring the operating parameters of the reader. The parameter configurator interface is shown in the following figure:



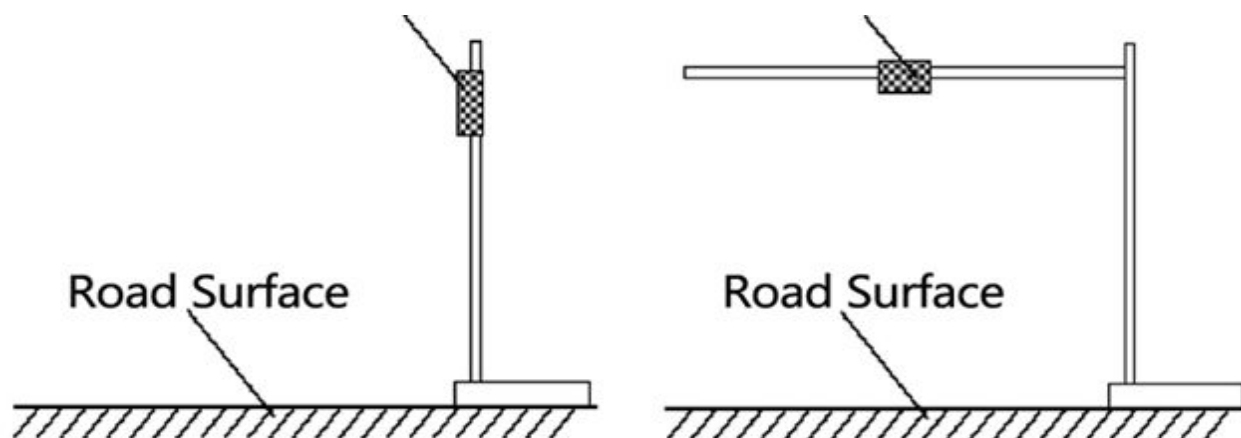
For the use of DEMO software, please refer to the "UHF Setup Software Manual"

Fourth, the installation of the reader

1. The installation mode of the reader

There are two types of reader stand installation methods: "side 1-line installation" and "L vertical type installation" see the following figure: the installation method can be selected according to the application

requirements and the actual situation on site. In general, the side installation read and write distance is closer, but the installation is simple; The top-loading method has a longer read-write distance, but the installation is complicated.



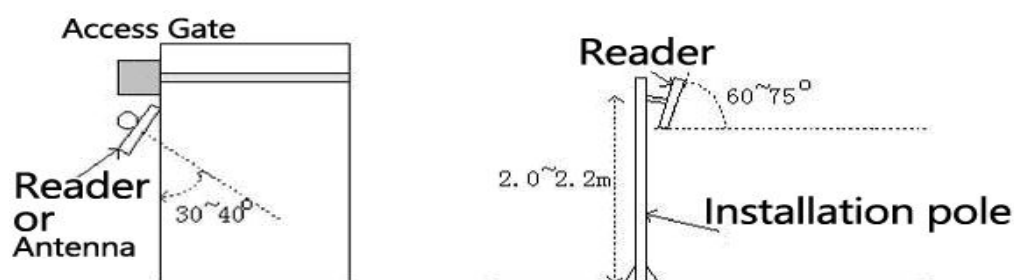
Reader side 1 type installation Reader L stand-up type mounting

2. Fixation and height adjustment of the reader

When the figure-of-one stand is mounted on the side, the installation rod diameter of the R16-12DB reader is required to be 40~80mm and the length 2.2m, it is best to use stainless steel material with a wall thickness greater than 1.2mm. Use the self-contained fasteners in the R16-12DB reader box to secure the reader to the top of the pole. Adjust the height of the R16-12DB reader center to the lane level according to the vehicle type (mainly large cars and small cars), which is generally about 2.0m (1.8~2.2m).

When the L-shaped stand is installed at the top, it is required to install the diameter of the L-shaped frame pole (or gantry shape) of 60 to 80mm and the diameter of the crossbar 40 to 80mm, it is best to use a stainless steel material with a wall thickness greater than 1.2 ~ 2.0 mm. The R16-12DB reader is also fixed to the crossbar near the middle of the lane using the self-contained fasteners in the box. The height of the crossbar from the ground is adjusted between 3.5m and 4.0m according to the needs of the vehicle height.

3. Reader azimuth adjustment



Antenna pitch angle: refers to the angle between the antenna tilting to the ground and the horizontal line, about 60 ~ 75 °

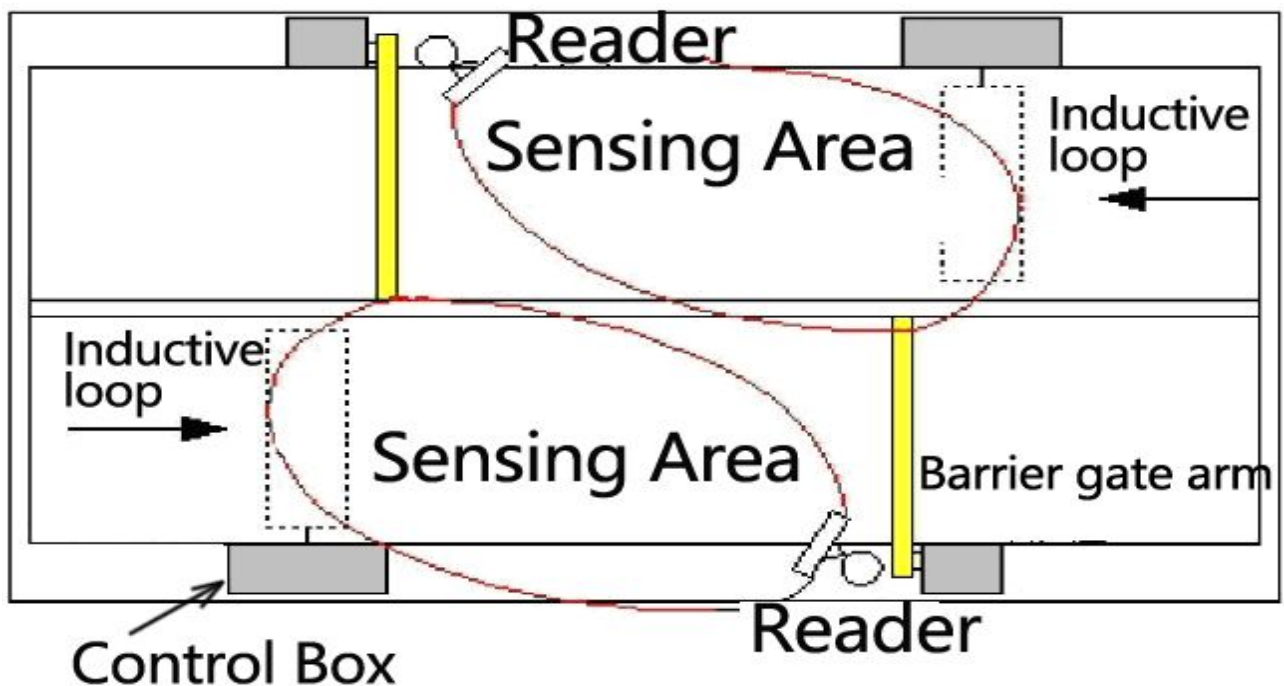
Antenna azimuth: Refers to the offset angle of the antenna in the direction of the lane, about 30 to 40 °

4. Installation example - vehicle parking lot management

The principles for choosing the installation location of the R16-12DB reader system are:

- 1) The R16-12DB reader should not be more than 1 meter away from the gate in a straight line
- 2) There is no item occlusion between the location of the R16-12DB reader and the tag card
- 3) R16-12DB reader and control equipment (or PC) as close as possible, and the use of shielded communication cable specific site installation implementation is generally determined according to the site conditions Explained below.
 - a) On-site installation method 1: there is no intermediate isolation of the road safety island, road control equipment (barriers) are installed on both sides of the road, and vehicles pass through the card reading area at a speed of less than 30 km / h.

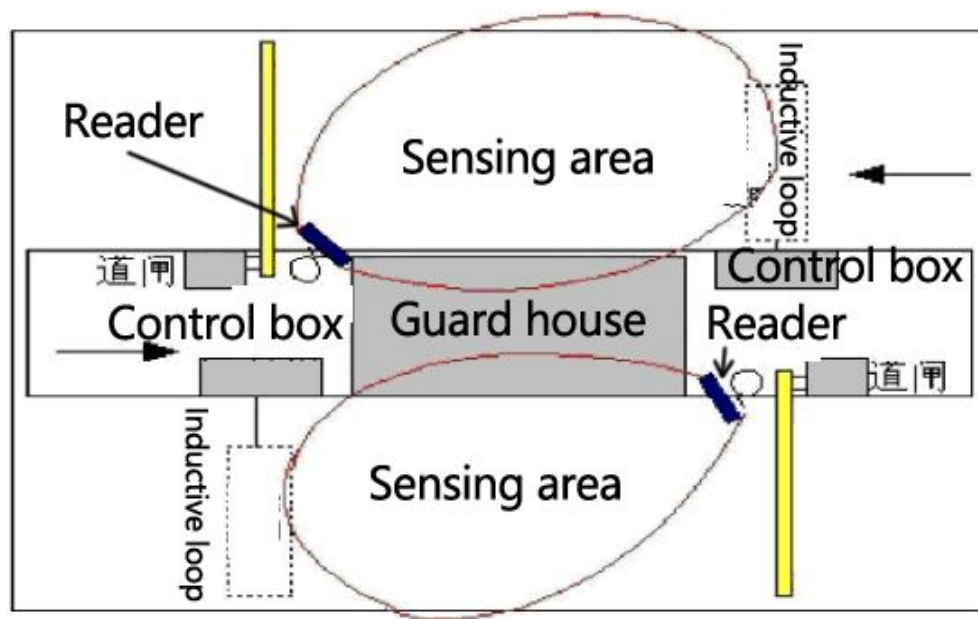
In this case: it is required that the reader (antenna) should be close to the barrier device, and at the same time, the effective range of its reading tag (the farthest straight-line distance is 0.5 meters to 35 meters) can cover the inlet ground sense coil or the outlet ground sense coil at the entrance and exit, as shown in the following figure.



- b) On-site installation method two: the road has an intermediate isolation of the safety island, the control equipment (gate) is installed on the middle isolation of the safety island, the vehicle speed is less than 10 km / h through the card reading area.

In this case: the R16-12DB reader is required to be close to the barrier device, and at the same time, the effective range of the tag reading (the farthest straight line distance is 0.5 meters to 35 meters) can cover

the inlet ground sense coil or the outlet ground sense coil at the entrance and exit, as shown in the following figure.



V. Secondary development of R16-12DB

Users can re-develop the reader application software as needed. We provide C#, VC, VB, Delphi, Java and other invocation routines, and provide WINCE, LINUX-driven development, please refer to the SDK Development Guide for the use of development kits